Transitioning this forensic analysis into a production pipeline requires a strict data architecture to prevent the ingestion of the synthetic and corrupted data exposed during the investigation.

Production Pipeline Architecture

Pipeline Layer	**Purpose & Operational Logic**	**Primary Output**
Raw	Unaltered ingestion of third-party APIs, webhooks, and vendor CSVs.	`raw_payments`, `raw_vendor_logs`
Staging	Standardization layer. Timestamps converted to UTC, basic deduplication (stripping webhook retries), and data type casting.	`stg_payments_utc`
Clean	Entity resolution and data cleansing. Dropping orphaned accounts, mapping redundant disposition codes (`PTP` to `PROMISE_TO_PAY`), and deduping borrowers.	`clean_master_accounts`
Golden	The validated, chronologically sound single source of truth for all business events and entities.	`golden_unified_ledger`
Feature	Application of business logic and time-window calculations (e.g., calculating `time_to_pay_hours` via backward merge).	`feat_attribution_matrix`

Metrics	Aggregated KPIs filtered by the feature logic, ready for BI consumption.	metric_72hr_channel_roi
Dashboard	The stakeholder visualization layer in Power BI/Tableau (The Investment Matrix).	Executive Dashboard

Engineering & Implementation Specifications

Data Contracts & Primary Keys

Enforce strict schema validation upon ingestion. If a record violates the contract (e.g., an account arrives without a valid borrower_id foreign key), it is immediately routed to a dead-letter queue, preventing "ghost accounts" from inflating the baseline. Every table must have a guaranteed unique identifier (e.g., payment_id, interaction_id).

Metric Definitions & Data Lineage

Metrics must be defined strictly in code, not in the BI tool. "Operational Revenue" is rigidly defined as amount where time_to_pay_hours <= 72. Data lineage must be fully mapped via a tool like dbt, ensuring leadership can trace the "72-Hour ROI" metric on the dashboard all the way back to the raw vendor API payload.

Incremental Processing & Backfills

To manage computing costs, the pipeline must process only net-new or modified records daily (incremental processing) rather than dropping and reloading the entire database. If leadership decides to test a 48-hour attribution window instead of 72 hours, a historical backfill pipeline must be triggered to recalculate all legacy feat_attribution_matrix rows using the new logic.

Late-Arriving Data

To solve the "August volume cliff" cohort bias, the metrics layer must implement a maturity threshold. Cohorts are flagged as "Incomplete" and excluded from performance dashboards until they reach at least 30 days of data maturity.

Data-Quality Checks & Anomaly Detection

Implement automated circuit breakers on the vendor data. If the staging layer detects a mathematically uniform distribution across telephony dispositions (e.g., PAID volume perfectly matches WRONG_NUMBER volume), the pipeline halts and alerts engineering of synthetic data fraud before it corrupts the executive dashboard.

Monitoring

Deploy infrastructure monitoring to track pipeline SLAs, data freshness, and row counts. If the clean_payments row count drops by 50% compared to the 7-day trailing average, alerting systems must page the data engineering team to investigate upstream ingestion failures.